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PROJECT TITLE: ARCHIVERSE — A CENTRALIZED PROJECT TRACKING AND CLIENT ENGAGEMENT SYSTEM FOR
ASAMOAH WELBECK & ASSOCIATES

INDIVIDUAL REPOSITORY: https://github.com/sarkodieshanell/personal_portfolio.git

TEAM REPOSITORY: <https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-44-jaskon-tech.git>

USER STORY: As a client, I want to submit a complaint regarding a project or service so that the architectural firm can review my concern, respond promptly, and improve customer satisfaction.

GITHUB ISSUE LINK: <https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-44-jaskon-tech/issues/39#issue-4989274575>

PRIORITY FEATURE: Submit a complaint.

User Role: Client

Goal: To submit a complaint about a project, service, staff interaction, or any issue experienced while working with the firm.

Value: This enables clients to communicate concerns directly to the firm, ensures issues are documented and tracked, promotes transparency, and improves the quality of service provided by Asamoah Welbeck & Associates.

Why is the operation needed?

In an architectural firm, clients may experience issues such as project delays, communication problems, incorrect drawings, billing disputes, or dissatisfaction with completed work. Without a formal complaint submission process, these issues may be ignored, forgotten, or handled inconsistently.

The Submit Complaint feature provides a centralized way for clients to report concerns. It ensures every complaint is recorded, assigned for review, monitored until resolution, and stored for future reference. This improves accountability, client trust, and overall service quality.

Caller and Provider components.

Caller Components

Client Portal: The client logs into Asamoah Welbeck & Associates, navigates to the Complaints section, completes the complaint form, and submits it.

Provider Component

Complaint Management Service: The backend:

- Validates the submitted complaint.
- Stores the complaint in the database.
- Generates a complaint reference number.
- Notifies the appropriate administrator or project manager.
- Updates the complaint status for tracking.

API/Interface Contract

Resource name: Submit a complaint.

HTTP Method: POST

Endpoint path: POST /api/complaints/

Path Parameters: None

Query Parameters: None

Headers

Authorization: Bearer <JWT Token>

Content-Type: application/json

Authentication and Authorization

Authentication: The user must be logged in and authentication is performed using a valid JWT Bearer Token.

Authorization: Only users with the Client role are permitted to submit complaints. Administrators and project managers may view and manage complaints but cannot submit complaints on behalf of clients through this endpoint.

Validation Rules

Basic Validation

- Subject cannot be empty.
- Description must be at least 20 characters.
- Project ID must be a valid integer.
- Category must be from the predefined list.
- Attachment (if provided) must be a valid type and size.

Business Rules

- Only authenticated clients may submit complaints.
- Clients can only complain about their own projects.
- The project must exist and be eligible for complaints.
- Duplicate complaints are prevented within 24 hours.
- Each complaint receives a unique reference number and a default status of Pending.

Success Response

HTTP Status Code: 201 Created

Meaning: The complaint has been successfully validated, saved in the database, assigned a unique complaint reference number, and is ready for review by the architectural firm's staff.

Response Field

Response Field	Explanation
complaint_id	Unique identifier of the complaint in the database.
reference_number	A tracking number given to the client for future enquiries.
status	The current state of the complaint (e.g., Pending, In Progress, Resolved).
message	A confirmation message indicating the outcome of the request.

User Interface Display

- Display a success notification, for example: "Complaint submitted successfully".
- Show the complaint reference number (e.g., CMP-2026-00105) so the client can save it for future reference.
- Display the complaint's current status as Pending.
- Redirect the client to the Complaint Details page or the My Complaints page, where they can view the complaint they submitted, monitor its status (e.g., Pending, In Progress, Resolved), review the complaint details and receive future updates from the architectural firm.

Error Model

Error Case	HTTP Status	Stable Error Code	User-Facing Message	Expected logging
User is not authenticated	401 Unauthorized	AUTH_REQUIRED	"Please log in to submit a complaint."	Log the failed request with timestamp, IP address, and reason: missing or invalid authentication token.
User is authenticated but is not authorised to submit complaints (e.g., Admin or Staff)	403 Forbidden	ACCESS_DENIED	"You do not have permission to submit a complaint."	Log the user's ID, role, attempted action, and timestamp
Validation fails (e.g., missing subject or invalid project ID)	400 Bad Request	INVALID_COMPLAINT_DATA	"Please correct the highlighted fields and try again."	Log the validation errors, request ID, and timestamp for troubleshooting.
Database or storage failure while saving the complaint	500 Internal Server Error	DATABASE_ERROR	"We couldn't submit your complaint at the moment. Please try again later	Log the database exception, stack trace, request ID, and timestamp. Alert the system administrator if appropriate.

Duplicate and Retry Behavior

Requirement	Description
Duplicate behavior	Duplicate submissions complaints are detected and rejected.
Duplicate check	Compare Client ID, Project ID, Subject, and submission time (within 24 hours).
Unique constraint	Each complaint has a unique <code>complaint_id</code> and <code>reference_number</code> .
Safe retry	After network interruptions or temporary server failures where no complaint was created.
Unsafe retry	After a successful <code>201 Created</code> response or when the complaint already exists.

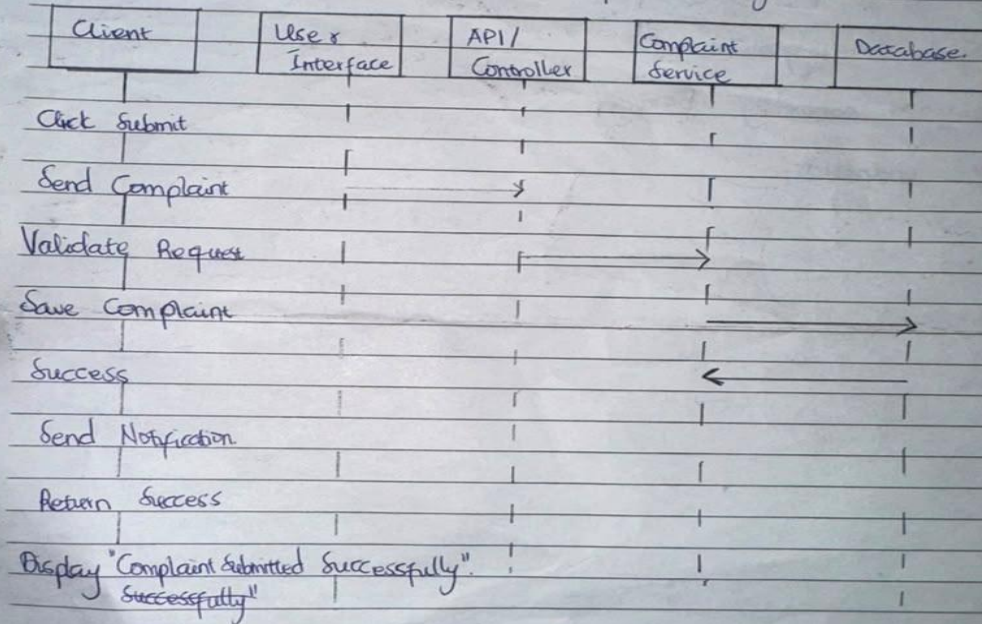
Diagram

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Date: 26 July, 2026.

Submit Complaint - Sequence Diagram.



Failure Point.

If Validation Fails, API → UI: Error Message Displayed.
"Unable to submit complaint". Please try again.

AI Disclosure

AI was used to help structure the API documentation, generate example test cases, and improve the wording and organization of the report. The final work was reviewed and adapted to fit the ARCHIVERSE project requirements.

